Predictive UPF Steering
Simulation, Forecasting, Optimization & Demo

Implementation review and evidence-backed technical presentation

Prepared for academic review and the C-DOT team · 09 August 2026

01
SIMULATE
30 s causal cohorts

02
FORECAST
10–80 min demand

03
OPTIMIZE
2 h cohort MPC

04
STEER
new sessions only

05
DEMONSTRATE
live operator view

All performance evidence in this deck is synthetic and artifact-backed.

SYNTHETIC DATA

The system works end to end—and the evidence boundary is explicit

A complete synthetic loop is implemented; the controller is a working demo candidate, not a production release.

TRAINING CORPUS
1.55M
group × 10-minute observations

FORECAST WAPE
7.63%
macro held-out, 10–80 min

MEAN-PAIR UL GAIN
10.52%
30 matched static/MPC pairs

SEVERITY-WEIGHTED GAIN
2.84%
tail risk still material

What is implemented

- **Deterministic synthetic data factory**
  30-second cohort simulation, 8 zones, 12 services, 24 UPFs, seeded surges and faults.
- **Trained probabilistic forecast bundle**
  2,304 direct calendar-ridge models with p50/p90/p95 conformal envelopes.
- **Causal cohort-state MPC**
  Twelve 10-minute windows, persistent-session state, static anchoring, same-state certificate and fallback.
- **Working dashboard demo**
  FastAPI + React, REST/WebSocket snapshots, class telemetry, per-UPF routing proof and frozen evidence.

Decision boundary

The accepted claim is reduced modeled overload exposure for future sessions—not guaranteed overload prevention.

- Worst paired scenario: −23.50%
- No established-session migration
- No live SMF/EMS actuation
- C-DOT telemetry and capacity calibration remain external

SYNTHETIC DATA

Review basis: implementation, documentation, artifacts and executable checks

01 **Implementation**
Simulator, forecasting, optimization, steering, API and React UI traced to source.

02 **Design records**
Architecture decisions, runbooks, traffic specification and C-DOT gap analysis reconciled.

03 **Frozen results**
Forecast bundle, optimizer tuning, oracle bound and 30-pair campaign evidence reviewed.

90 / 90
backend tests passed

- contracts and telemetry reconstruction
- deterministic simulator and Parquet
- forecast leakage and bundle integrity
- HiGHS, cohort MPC and oracle bounds
- API, story, rewind and campaign evidence

6 / 6
demo preflight gates passed

- traffic registry
- 3-UPF / 6-class scenario
- forecast bundle checksum
- FastAPI OpenAPI: 20 paths
- production-built operator console
- frontend bundle checksum

Evidence rule

Synthetic values, workstation benchmarks and external integration assumptions are never mixed into a production claim.

The worktree is currently dirty; frozen artifacts preserve hashes and exact code identities where documented.

SYNTHETIC DATA

The implementation is an artifact-backed, causal closed loop

SCENARIO
Seeded manifest
+ topology + events

→

SIMULATE
30 s cohorts
+ traffic accounting

→

AGGREGATE
20 ticks →
10 min observations

→

FORECAST
p50 / p90 / p95
1–12 windows

→

OPTIMIZE
Cohort-state MPC
+ static certificate

→

ACTUATE
Weighted
rendezvous
new sessions only

OFFLINE — LIVE

Artifact factory

PBS or workstation shards publish immutable Parquet, JSONL, selection audits and metadata hashes. The cluster is not required during a demo.

Control decision

At a decision boundary, only closed history and declared known-at events are visible. The first MPC action is compared with static from the identical cohort state.

Presentation runtime

One FastAPI process advances the simulator, emits ordered WebSocket deltas, serves the React console and retains an append-only audit trail.

SYNTHETIC DATA

Traffic is modeled by controllable groups—not by anonymous aggregate load

Population and dimensions

ZONES

8

urban · industrial · airport · stadium · rural

SERVICES

12

consumer, enterprise, critical and IoT

GROUPS

96

8 zones × 12 service profiles

ELIGIBLE UPFS

6

2 edge + 2 regional + 2 central

Forecast / QoS key

`(zone, DNN, S-NSSAI, 5QI)`

5QI is descriptive in v1; it is not assumed to be an independent SMF selection key.

Zone and service diversity

- north-urban
- south-urban
- east-urban
- west-urban
- industrial
- airport
- stadium
- rural

Why grouping matters

- Forecasts remain local to an operationally meaningful class.
- UPF eligibility and path latency are enforced per group.
- Routing weights can shift one class without assuming every session can reach every UPF.

SYNTHETIC DATA

The executable stochastic model is simple, reproducible and deliberately bounded

$$N(g,t) \sim \text{Poisson}(\lambda_g \times F_{g,t})$$

$F_{g,t}$ = daily profile × weekend factor × U(0.86, 1.16) weekly noise × active surge

Factors update hourly; arrivals are drawn every 30 seconds.

Determinism by construction

- manifest seed fixes events
- SHA-256-derived random stream per group
- independent arrival and lifetime streams
- same seed + policy ⇒ same logical run

SESSION LIFETIME

Discrete uniform integer draw within each class range.

SESSION RATE

Fixed UL/DL Mbps per active session for a class.

STATE

Placed sessions become aggregate cohorts with explicit departure steps.

LOAD

Offered demand is the sum of fixed rates for all active sessions.

Not implemented in the extreme generator: AR(1), Markov bursts, heavy-tailed rate/lifetime, persistent UE identity, mobility, packet-level traffic or telemetry faults.

SYNTHETIC DATA

Twelve service classes create distinct temporal and directional demand

Values are per active session; lifetime ranges are from the executable extreme profile.

SERVICE / DNN	5QI	ARRIVALS / 30 s / ZONE	LIFETIME	UL / DL Mbps	CALENDAR
Consumer video	9	201.6–403.2	1–12 h	0.15 / 4.00	Evening
Social / live	8	112.0–224.0	10 m–2 h	2.00 / 1.00	Evening
Gaming	3	134.4–268.8	40 m–6 h	0.20 / 0.40	Late
IMS voice	1	89.6–179.2	10 m–4 h	0.08 / 0.08	Commute
Enterprise	7	112.0–224.0	30 m–8 h	1.00 / 2.00	Business
Video conference	2	89.6–179.2	30 m–6 h	1.50 / 1.50	Business
Industrial URLLC	82	67.2–134.4	4–24 h	0.10 / 0.08	Industrial
Massive IoT	9	224.0–448.0	6–48 h	0.020 / 0.004	Flat
Connected vehicle	84	112.0–224.0	5 m–1 h	0.50 / 1.00	Commute
Edge AI	7	56.0–112.0	10 m–2 h	3.00 / 6.00	Business
Cloud backup	9	22.4–44.8	1–12 h	8.00 / 2.00	Overnight
Public safety	65	33.6–67.2	20 m–3 h20	2.00 / 2.00	Flat

The same arrival count drives sessions, UL and DL via fixed class rates; identical target WAPE does not imply three independent traffic processes.

SYNTHETIC DATA

Scenario generation covers normal regimes, random surges and network disturbances

DAILY / WEEKLY	258,048 factors	service curve × weekend × group/week noise	demand seasonality
FLASH CROWDS	192 episodes	12/week · random group · 2–10 h · ×2.5–8.0	localized step surges
BROWNOUTS	128 episodes	8/week · 1–8 h · UL ×0.18–0.70	capacity, queue, drops
NEAR OUTAGES	48 episodes	3/week · 1–4 h · degraded + 1% capacity	severe residual stress
LATENCY	80 episodes	5/week · 1–6 h · +25–140 ms	path-cost disturbance

Surge IDs are not stored in run.parquet; their multipliers are folded into hourly arrival-factor events. Fault effects appear as resulting UPF state.

SYNTHETIC DATA

Eligibility and directional safety envelopes constrain every placement

24-UPF logical hierarchy

One traffic group sees exactly six eligible destinations

ZONE

north · south · east · west

industrial · airport · stadium · rural

EDGE — **2 per zone** →

REGIONAL — **4 total** →

CENTRAL — **4 total** →

Per-instance capacity envelope

TIER	COUNT	UL	DL	SESSION CAP	SAFE
Edge A	8	480 G	1.44 T	1.44 M	78% / 82%
Edge B	8	560 G	1.28 T	1.44 M	78% / 82%
Regional	4	1.12 T	2.40 T	3.52 M	80% / 85%
Central	4	1.92 T	3.84 T	5.12 M	75% / 85%
TOTAL	24	20.48 T	46.72 T	57.60 M	mixed

Admission filters: health ∊ {healthy, degraded} · group eligible · locality path exists · optional max latency · session capacity available.

SYNTHETIC DATA

Each 30-second tick preserves causal order and all traffic accounting states

1 **CLOSE** At a 10-minute boundary, close the prior 20-tick history bucket.

2 **EVENTS** Apply arrival, capacity, health and latency events for this tick.

3 **POLICY** Replan if due using current state and only causal history.

4 **ARRIVALS** Draw exact Poisson arrivals per group from independent streams.

5 **SELECT** Filter eligible/healthy UPFs and run weighted rendezvous per session.

6 **ADMIT** Check session capacity; draw uniform lifetime and create a cohort.

7 **SERVE** Apply capacity, queues and drops separately for UL and DL.

8 **EMIT** Write step telemetry, audit samples and boundary group/UPF state.

Traffic is recorded as offered → carried / queued / dropped, while admission failures and unplaced rejections remain separate.

SYNTHETIC DATA

The 16-week extreme campaign is large enough to stress the full offline path

DURATION
112 d
322,560 × 30-second ticks

TRAINING ROWS
1.548 M
96 groups × 16,128 buckets

PROJECTED SESSIONS
4.39 B
routed arrivals; audit sampled 1:5,000

ARTIFACTS
≈9.6 GiB
one 112-day extreme shard

ARTIFACT	GRANULARITY	WHAT IT PROVES
run.parquet	1 row / 30 s	group arrivals + nested per-UPF UL/DL/session state
group_upf_buckets[]	10 min boundary	joint class/UPF active, admitted and rejected state
selection-audits.parquet	1 / 5,000 sessions	hashed key, eligible set, weights and selected UPF
run.jsonl	readable adapter	metadata + steps + retained selection audits
metadata.json	1 / shard	hashes, source identity, host/job and aggregate metrics
forecast bundle	1 immutable JSON	2,304 models, metrics, calibration and checksum

Measured one-day calibration: 5:06 wall time · 588 MiB peak RSS · 87.6 MiB output. A 112-day shard projects to ~9 h 32 m and ~64 GiB retained memory.

SYNTHETIC DATA

Training converts canonical 30-second Parquet into leakage-safe direct forecasts

20 TICKS
sum group arrivals

→

10 MIN
mean UPF residual state

→

OBSERVATION
sessions + UL + DL

→

DIRECT MODELS
3 targets × 8 horizons

→

CONFORMAL
p50 + p90 / p95 widths

→

BUNDLE
checksum + metadata

Exact training row

- new_session_count = Σ arrivals in the 20 ticks
- new UL/DL = count × fixed class Mbps/session
- residual sessions and offered Mbps = duration mean across the bucket
- one DemandObservation emitted per controllable group

What the v1 trainer does not consume

- queue, drop or rejection fields as targets
- selection audits or explicit event IDs
- packet counters, CPU, memory or resets/gaps
- per-group measured carried throughput

SYNTHETIC DATA

The chosen forecaster favors transparency, speed and numerical stability

Calendar-ridge direct multi-horizon model

FEATURE	ROLE
last value	autoregressive level
rolling mean (6)	one-hour causal baseline
recent trend	local direction
daily seasonal lag	same-time prior-day structure
sin/cos time of day	continuous daily phase
sin/cos day of week	weekly calendar phase
intercept	group/horizon baseline

96 groups × 3 targets × 8 horizons = 2,304 nine-feature ridge systems

Engineering optimizations

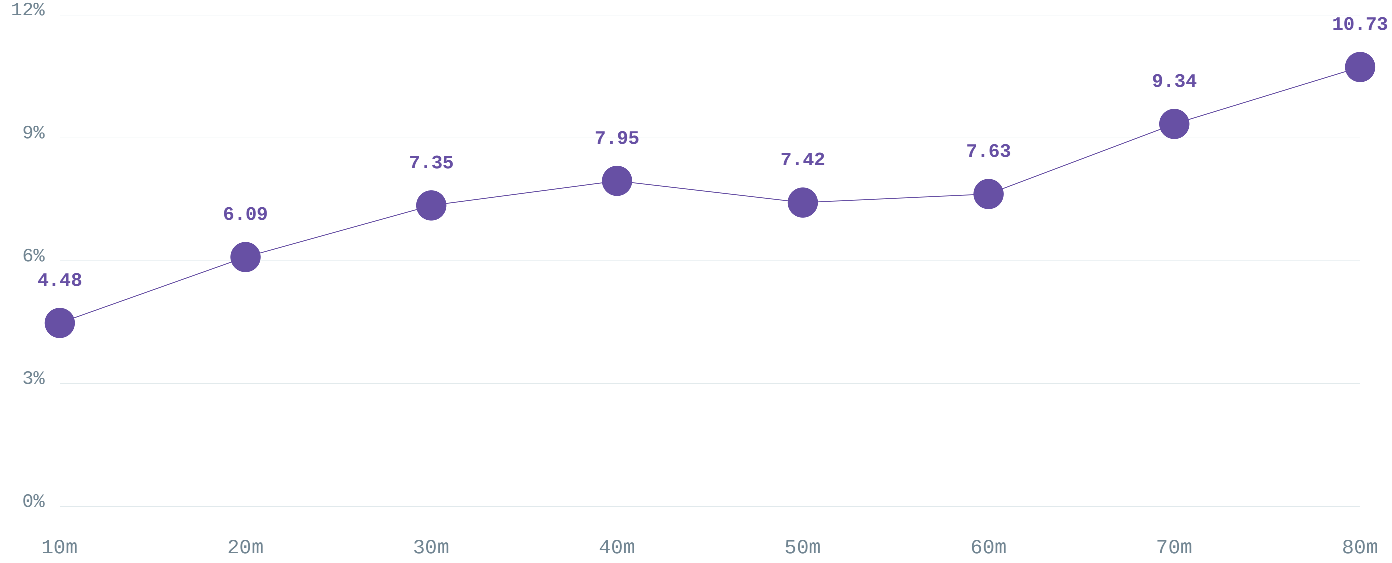
- **Vectorized row construction**
  NumPy builds each sequence in O(n); the prior scalar path rebuilt full histories.
- **RMS feature scaling**
  stabilizes national-scale normal equations while preserving the bundle coefficient contract.
- **Ordered split**
  70% train / 15% calibration / 15% test; features never cross the forecast origin.
- **Median bias correction**
  calibration residual median shifts the non-negative p50 point forecast.
- **Split conformal + ACI**
  residual widths provide p90/p95; adaptive alpha reacts to realized coverage.

3:03 total · ~13 s fitting · ~1.8 GiB peak RSS · GPU unnecessary for v1

SYNTHETIC DATA

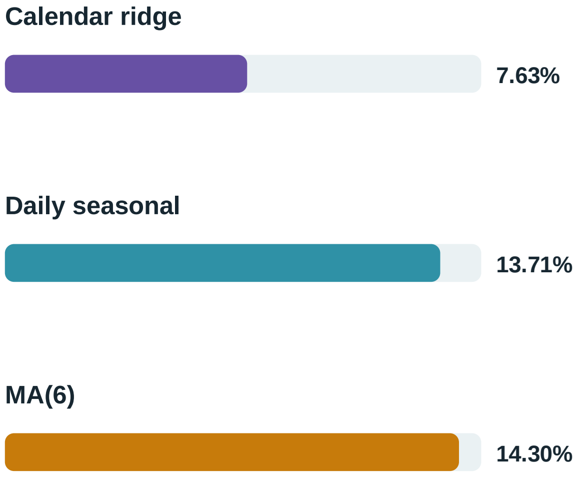
Held-out performance is strong at operational horizons and degrades predictably with lead time

Macro WAPE by horizon



p90 upper-bound coverage stays between 93.10% and 94.96% across all eight horizons.

Fair baseline comparison



44.36% lower WAPE than daily seasonal naive.

SYNTHETIC DATA

Surges—not brownouts—are the dominant forecast weakness

REGIME	WAPE	P90 COVERAGE	INTERPRETATION
Normal	6.51%	94.19%	well calibrated
Surge	11.08%	30.73%	unannounced step-change is not predictable beforehand
Brownout	5.09%	87.88%	network state changes; offered demand does not
Near outage	5.29%	73.96%	demand accuracy stays normal; coverage weakens
Latency incident	5.04%	92.36%	path state changes, not demand

Operational consequence

The controller needs robust fallback and static-relative certification; forecast accuracy alone cannot guarantee benefit.

Evaluation caveat

The v1 70/15/15 split differs slightly from the manifest's explicit 11/2/3-week boundary; event-stratified release reporting remains incomplete.

SYNTHETIC DATA

The optimizer evolved only after the evidence disproved the first design

01 · ONE-WINDOW LP

p95 forecast + residual load

Validation failed

→

02 · MECHANISM TUNING

hints · anomaly · lifetime · caps

Still worse than static

→

03 · ORACLE BOUND

full-day cohort relaxation

100% modeled UL reduction possible

→

04 · COHORT MPC

12-window state transition

10.52% mean-pair gain

Core lesson

New-session decisions persist for hours. A controller that optimizes only the next bucket can create residual concentrations that it cannot later undo.

The successful redesign carried cohort survival through a two-hour horizon and compared every candidate with static from the same state.

STAGE 3 · FAILURE ANALYSIS

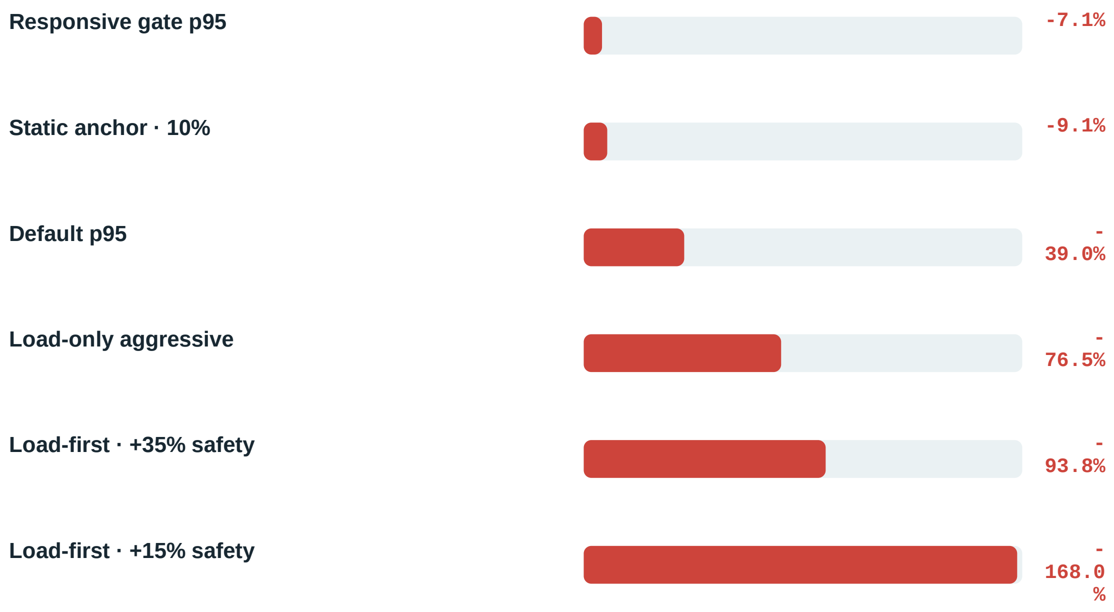
SYNTHETIC DATA

Why the one-window HiGHS controller lost to a strong static baseline

Mechanism of failure

- **Myopic horizon**
  Minimizing next-window peak utilization ignores the future occupancy of long-lived cohorts.
- **Persistent placement**
  Only future sessions are controllable; already placed sessions cannot be rebalanced.
- **Correlated concentration**
  A per-group weight cap does not prevent many groups from choosing the same failure domain.
- **Forecast step-changes**
  Surge p90 coverage collapses before an unannounced jump becomes observable.
- **Static is robust**
  Capacity-weighted rendezvous spreads each group broadly and continuously.

Validation outcome versus static



No profile passed guardrails; reserved fresh test seeds were preserved rather than used for selection.

SYNTHETIC DATA

Cohort-state MPC plans the consequences of today's weights across twelve windows

State transition model

$$\text{Projected load}(u,t) = \text{anchored cohorts}(u,t) + \Sigma_{g,\tau\le t}\ x(g,\tau,u) \cdot \text{demand}(g,\tau) \cdot \text{survival}(g,t-\tau)$$

HORIZON	12 × 10 min = 2 hours	DEMAND	causal MA(6) p95; scheduled multipliers only after known_at_step
STATE	exact active cohorts by group, UPF, remaining lifetime and directional rate	CAPACITY PATH	current state + declared future health/capacity events
DECISION	continuous group→UPF weights for each horizon window	ACTION	only the first window is published; replan every 10 minutes

Frozen demo profile

MAX UPF WEIGHT	0.75
STATIC BLEND	50%
SOLVER TIMEOUT	2.0 s
OVERLOAD COST	1×
PHYSICAL DROP COST	10×
TERMINAL EXPOSURE	1×
UNPLANNED CAPACITY	fallback static

SYNTHETIC DATA

Every MPC action is guarded by a same-state static certificate

1. **FORECAST** — 12-window demand
2. **SOLVE** — HiGHS LP
3. **REPLAY** — static from same state
4. **CERTIFY** — guardrails + gain
5. **PUBLISH** — or exact static

Certificate requires

- UL overload area no worse than static
- DL overload area no worse than static
- session overload no worse than static
- UL and DL physical drops no worse than static
- minimum score and modeled UL improvement

Fallback triggers

- insufficient history or forecast error
- solver infeasible / error / timeout
- certificate rejects the candidate
- observed unplanned capacity state
- no healthy eligible route exists

SYNTHETIC DATA

The frozen 30-pair campaign passes the working-demo gate

MEAN-PAIR UL

+10.52%

bootstrap 95% CI: 4.81–16.93%

SEVERITY-WEIGHTED UL

+2.84%

dominated by highest-overload days

UL DROPPED BYTES

+12.42%

aggregate reduction

DL DROPPED BYTES

+9.34%

aggregate reduction

Exactly paired experiment

- **30 fresh seeds**
  34001–34030; one simulated day per pair.
- **Common random numbers**
  same scenario, arrivals, lifetimes, event schedule and rendezvous namespace.
- **Four stress families**
  8 surge · 8 scheduled fault · 7 unannounced outage · 7 mixed.
- **Aggregate guardrails**
  DL overload, both directional drops and establishment failures all pass.

Release decision

Working demo candidate—not production-ready.

WORST MATCHED PAIR

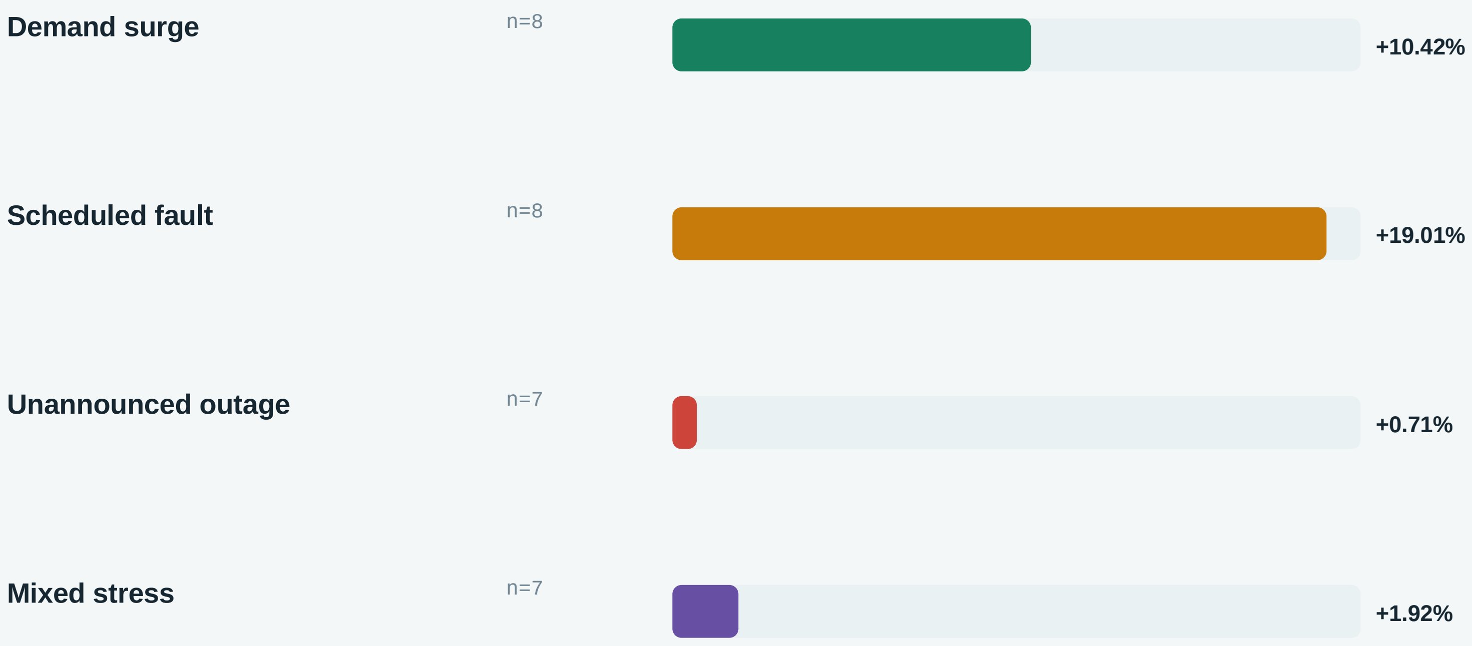
−23.50%

scheduled-fault tail regression

SYNTHETIC DATA

Average gains are broad, but robustness is weakest under surprise and fault tails

AGGREGATE UL OVERLOAD-AREA REDUCTION



Worst pair by scenario

Demand surge	+2.57%
Scheduled fault	-23.50%
Unannounced outage	-9.84%
Mixed stress	-8.28%

Next release gate: positive severity-weighted confidence bounds on untouched seeds, plus materially reduced unannounced-outage and scheduled-fault tails.

SYNTHETIC DATA

The dashboard is a real causal runtime—not a prerecorded animation

REACT OPERATOR CONSOLE
Live Dashboard · Evidence · Technical Detail · Expert mode

REST + WEBSOCKET
commands and snapshots · ordered versioned deltas · reconnect

FASTAPI ORCHESTRATOR
run lifecycle · auth roles · audit · story checkpoints · analytics

CAUSAL SIMULATION LOOP
30 s tick · 10 min control · forecast · MPC · actuator

FROZEN ARTIFACTS
scenario · forecast bundle · MPC profile · 30-pair evidence

The server chooses the policy before realizing the tick; event injection never rewrites prior telemetry. Rewind restores complete runtime state.

SYNTHETIC DATA

The operator sees both the decision and the evidence behind it

LIVE DASHBOARD

- per-UPF capacity, load, headroom and health
- new-session route widths and weight deltas
- forecast → certificate → apply/hold trace
- completed surge debrief and iteration ledger

EVIDENCE

- 30 matched-pair frozen campaign
- mean-pair CI and severity-weighted total
- scenario-level aggregate and worst pair
- artifact identity and production boundary

TECHNICAL DETAIL

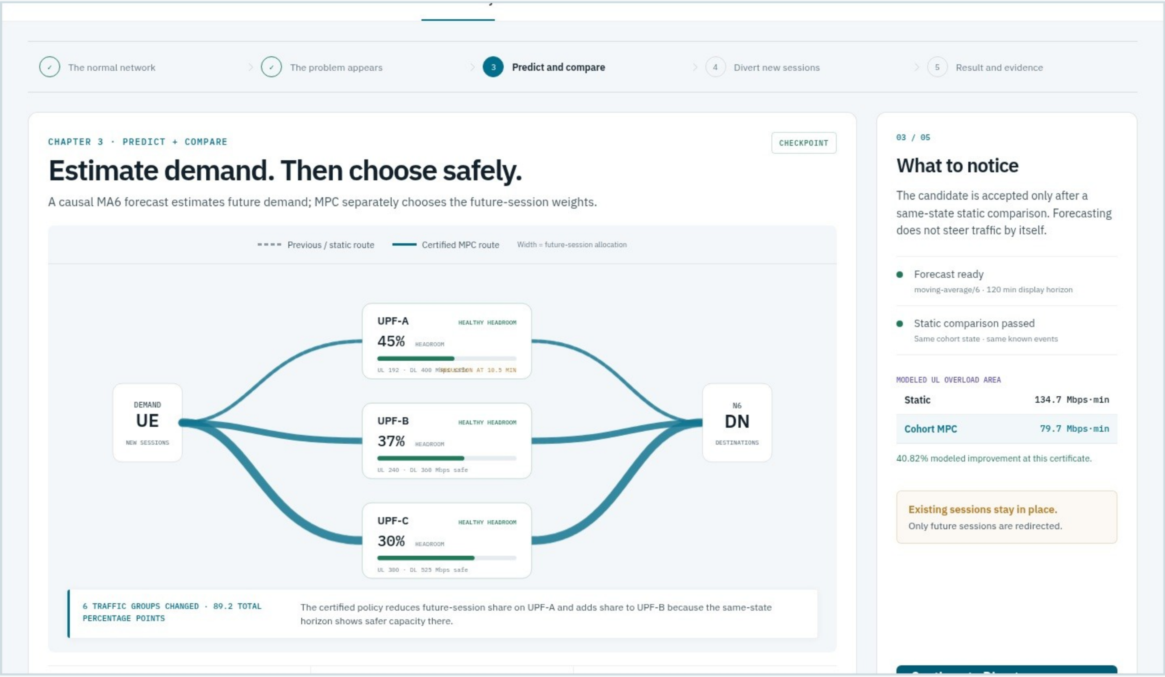
- class arrivals, admissions and rejections
- p50/p90/actual forecast table
- static vs MPC certificate metrics
- raw trace, deployment boundary and expert controls

DISPLAYED DATA SOURCES

simulator StepResult · canonical group/UPF buckets · Forecast · Policy · certificate · frozen campaign JSON

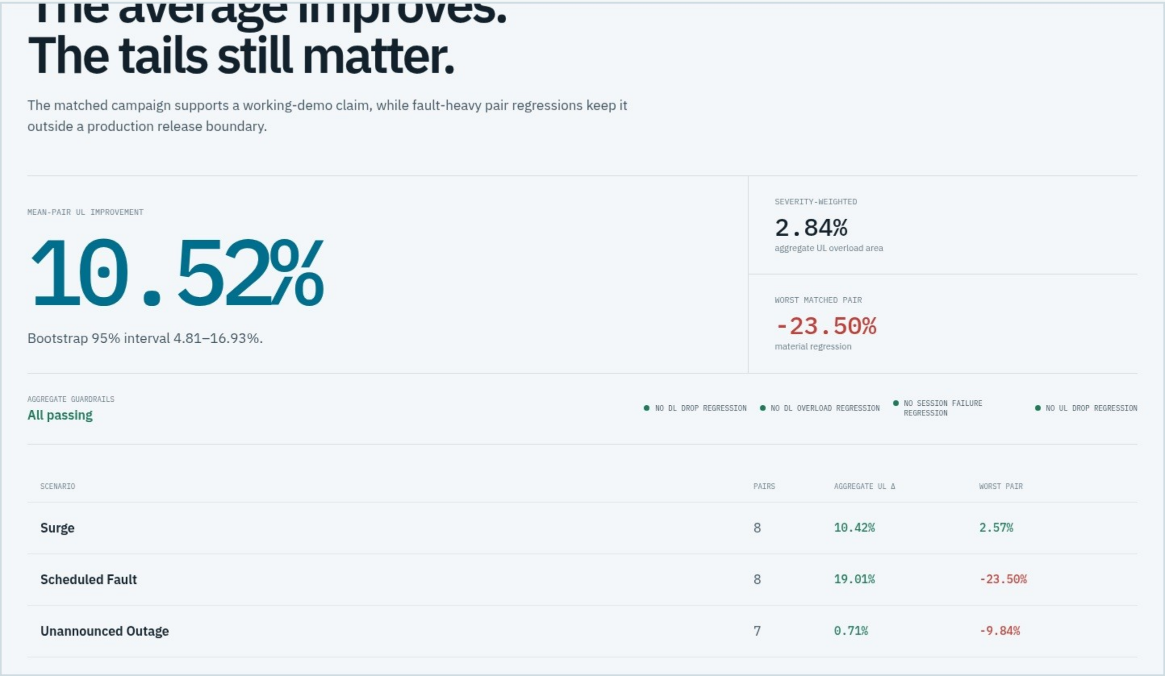
SYNTHETIC DATA

The actual console keeps routing, evidence and caveats visible



Predict + compare checkpoint

Route widths encode future-session allocation; the side rail exposes the same-state static comparison and the no-migration caveat.



Frozen campaign evidence

The evidence view leads with the 10.52% average but places severity weighting, worst pair and release status beside it.

SYNTHETIC DATA

Traffic diversion is deterministic weighted placement for newly arriving sessions

SIMULATOR PLACEMENT PATH

```
requested = policy_weights[group_id]
allowed = {u:w for u,w in requested.items()
           if u in eligible_upfs
           and state[u].health in {healthy,degraded}}
weights = normalize(allowed)

selected = rendezvous_select(
    session_key, stable_namespace, weights)

if active[selected] >= session_capacity:
    reject()
else:
    lifetime = randint(min_steps, max_steps)
    create_anchored_cohort(selected, lifetime)
```

The policy ID does not re-salt the hash namespace; paired runs share the selection randomization.

What a weight change means

- **Before**
  Each new session hashes across the current eligible weighted set.
- **Policy epoch**
  MPC publishes new normalized weights after certification.
- **After**
  Only later arrivals use the new weighted scores.
- **Persistence**
  Admitted sessions stay on the selected UPF until their lifetime ends.
- **No migration**
  Existing TEIDs/sessions are never moved by this code.

SYNTHETIC DATA

A five-minute guided story makes the control loop explainable

TIME	CHECKPOINT	WHAT THE AUDIENCE SEES	DEFENSIBLE CLAIM
0:00	Normal network	three UPFs inside safe envelopes	established sessions remain anchored
0:35	Problem appears	stadium demand rises; known UPF-A UL reduction	pressure is visible before the route changes
1:20	Predict + compare	causal MA6 forecast + same-state certificate	forecast does not steer traffic by itself
2:35	Divert new sessions	new route widths and per-UPF weight deltas	future arrivals shift; existing sessions do not
3:40	Result + evidence	realized outcome, loss and frozen 30-pair evidence	modeled exposure is reduced, not eliminated

Deterministic checkpoints are server-side state: browser reload, brief disconnect and rewind do not invent or recompute realized history.

SYNTHETIC DATA

Verification spans contracts, causality, optimization and the presentation surface

CONTRACTS
versioned schemas · round-trip fixtures · normalized policies

CAUSALITY
no future observations · known_at_step enforcement · append-only history

SIMULATOR
deterministic streams · queues/drops · Parquet schema · audit sampling

OPTIMIZER
hand-solvable LPs · eligibility · diversification · same-state certificate

EXPERIMENTS
exact pairing · reserved seeds · immutable outputs · bootstrap interval

DEMO
auth roles · OpenAPI · WebSocket order · rewind exactness · visual snapshots

BACKEND TESTS
90
all passed in 4.56 s

API PATHS
20
FastAPI preflight

VISUAL BASELINES
13
desktop, tablet and mobile

PREFLIGHT
PASS
all data labeled synthetic

SYNTHETIC DATA

Simulation and production integration are intentionally separated

Available now

- deterministic cohort simulation and event injection
- canonical Parquet, audit and hashed metadata
- trained forecast bundle with uncertainty bounds
- causal cohort MPC with static certificate
- simulated new-session actuation
- Prometheus-compatible synthetic metrics
- REST/WebSocket operator dashboard
- advisory and placeholder actuator interfaces

Requires C-DOT / testbed

- live Prometheus metric names, labels and scrape semantics
- measured UPF UL/DL/session capacity envelopes
- topology, locality and eligibility truth
- authenticated SMF/EMS policy publication contract
- confirmation of established-session relocation support
- free5GC/PacketRusher integration and saturation tests
- production drift, missingness and counter-reset behavior
- untouched release campaign and tail-risk gate

SYNTHETIC DATA

Recommended path from working demo to C-DOT-calibrated pilot

01 **Observe** — Map N3/N6 counters, sessions, health and label cardinality into versioned telemetry contracts.

02 **Calibrate** — Measure directional throughput/session envelopes and reproduce overload thresholds on target UPF builds.

03 **Replay** — Run the existing forecaster and MPC in advisory mode against recorded representative history.

04 **Shadow** — Publish recommendations to an external store without actuation; compare static and MPC decisions online.

05 **Pilot** — Enable bounded new-session steering with fail-closed rollback and explicit operational approval.

Go/no-go principle: calibrate data and authority first; keep the current synthetic evidence intact as a reproducible engineering baseline.

SYNTHETIC DATA

The engineering result is credible because the limitations are visible

Predict early.
Certify against static.
Steer only what is controllable.

The repository demonstrates an end-to-end synthetic control loop with measurable average benefit, reproducible artifacts and an honest production boundary.

FINAL TAKEAWAYS

- 7.63% forecast WAPE
- 10.52% mean-pair UL improvement
- 2.84% severity-weighted benefit
- all aggregate guardrails pass
- fault-tail risk remains
- new-session placement only

Discussion

What telemetry, capacity and actuation interfaces can C-DOT expose for a calibrated shadow pilot?

SYNTHETIC DATA

Appendix: metric definitions used in the experiments

METRIC	DEFINITION	WHY IT MATTERS
WAPE	Σ|actual−p50| / Σ|actual|	scale-normalized forecast point error
p90 coverage	fraction where actual ≤ predicted upper p90	uncertainty calibration for conservative planning
UL overload area	Σ max(0, load/safe_capacity−1) × seconds	severity × duration above safe envelope
Overload duration	seconds with load above safe capacity	duration only; ignores magnitude
Dropped bytes	overflow beyond physical capacity and bounded queue	modeled service loss
Establishment failures	sessions rejected by capacity or no eligible route	admission quality guardrail
Mean-pair reduction	mean of per-seed relative reductions	equal weight to each paired scenario
Severity-weighted reduction	reduction in total area across all pairs	weights heavy overload days more strongly
Total variation	½ Σ|new weight−old weight|	policy churn / amount of routing change

The primary supplied-demo metric is directional UL overload area; DL is reported separately and cannot be hidden by a combined average.

SYNTHETIC DATA

Appendix: source map for implementation and evidence review

STAGE	PRIMARY IMPLEMENTATION	PRIMARY EVIDENCE / DOCS
Scenario generation	experiments/build_extreme_history_manifest.py configs/extreme_training_profile.json	docs/extreme-data-spec-and-cdot-gap-analysis.md docs/traffic-model-spec.md
Simulator	simulator/macro/engine.py simulator/macro/config.py · model.py	docs/end-to-end-runbook.md docs/system-architecture-decisions.md
Forecaster	experiments/train_forecaster.py forecasting/bundle.py · baselines.py · metrics.py	docs/extreme-forecaster-v1-results.md
One-window LP	optimization/highs.py simulator/macro/controllers.py	docs/extreme-optimizer-tuning-results.md docs/static-controller-deep-research.md
Oracle + cohort MPC	optimization/oracle_bounds.py optimization/cohort_mpc.py	docs/extreme-oracle-bound-results.md docs/cohort-mpc-full-campaign-results.md
Steering	steering/hashing.py · policy.py · gate.py	docs/cdot-session-migration-decision.md
Working demo	demo_api/main.py · runtime.py · story.py frontend/src/App.tsx · views.tsx	docs/presenter-guide.md frontend/tests/demo.spec.ts-snapshots/

Frozen campaign evidence: demo_api/data/cohort_mpc_full_campaign_evidence_v1.json · source SHA-256 bd1d3727…b662c